Chapter 2:
Relational Data Model

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1. Introduction

- *Relational Data* Model is based on the concept of **relations**.
- *A relation* is a mathematical concept based on the formal foundation of set theory.
- This model was proposed by Dr. EF Codd in 1970.

2.1 Attribute

- Properties of an attribute:
  - ***Name*** : character string (mnemonic)
  - ***Data types*** : Number, String, Time, Logical, OLE.
  - ***Value domain*** : the set of values an attribute can take. The value domain of attribute A is denoted by ***Dom*** (A).
- **For example** : GIOITINH data type is String, value domain

  Dom(GIOITINH)=('Male','Female')

- If an attribute had no value or an unknown value

  => **Null** value

2.2 Relation

- **Definition** : A relation is a finite set of attributes.
  - **Symbol** : $Q(A_1, A_2, \ldots, A_n)$
  - Where:
    - Q is the name of the relationship,
    - $Q^+ = \{A_1, A_2, \ldots, A_n\}$ is the set of attributes of the relationship Q
  - **For example** :

HOCVIEN (Mahv, Hoten, Ngsinh, Gioitinh, Noisinh, Malop)

LOP (Malop, Tenlop, Siso, Trglop, Khoa)

2.3 Tuple

- **Definition :** A tuple is the information of an object belonging to a relation. It is called a record or row.
- A relation is a table with columns called attributes and each row is called a tuple.
- A tuple of a relation $Q(A_1, A_2, \ldots, A_n)$ is $q = (a_1, a_2, \ldots, a_n)$

  with $\forall\ a_i \in Dom\ (A_i)$
- **For example** : HOCVIEN(Mahv, Hoten, Ngsinh, Noisinh) has q=(1003,Nguyen Van Lam, 1/1/1987,Dong Nai) which means a student has ID 1003, full name is Nguyen Van Lam, born on 1/1/1987 in Dong Nai

2.4 Instance of relation

- **Definition** : An instance of a relation is a set of tuples of the relation at a point in time.
- **Notation:** the instance of the relation Q is T_Q
- **For example** : T_{HOCVIEN} is the current instance of the HOCVIEN relationship, with the following tuples:

HOCVIEN				
Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
K1103	Ha Duy Lap	Nam	Nghe An	K11
K1102	Tran Ngoc Han	Nu	Kien Giang	K11
K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11

2.5 Description (Tân từ)

- **Definition** : a rule used to describe a relationship.
- **Symbol** : ||Q||
- **Example** :

THI (Mahv, Mamh, Lanthi, Diem)

||THI||: each student is allowed to take a subject multiple times, each time the exam records which student took which subject? which exam? and what is the score?

2.6 Projection (1)

- **Projection** : Used to extract the value of some attributes in the list of attributes of a relation.
- **Notation** : the projection of relation R onto attribute set X is ***R[X] or R.X***.
- **For example** :

	HOCVIEN				
	Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
$hv_1 =$	K1103	Ha Duy Lap	Nam	Nghe An	K11
$hv_2 =$	K1102	Tran Ngoc Han	Nu	Kien Giang	K11
$hv_3 =$	K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11

2.6 Projection (2)

- Projection of the HOCVIEN relation onto the NoiSinh attribute of the HOCVIEN relation:

HOCHIEN[Noisinh] = {'Nghe An','Kien Giang','Tay Ninh'}

HOCVIEN				
Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
K1103	Ha Duy Lap	Nam	Nghe An	K11
K1102	Tran Ngoc Han	Nu	Kien Giang	K11
K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11

2.6 Projection (3)

- Projection onto a set of attributes X={Hoten,Noisinh} of the relationship HOCVIEN

HOCVIEN[Hoten, Noisinh] = {('Ha Duy Lap', 'Nghe An'), ('Tran Ngoc Han', 'Kien Giang'), ('Tran Ngoc Linh','Tay Ninh')}

HOCVIEN				
Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
K1103	Ha Duy Lap	Nam	Nghe An	K11
K1102	Tran Ngoc Han	Nu	Kien Giang	K11
K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11

2.6 Projection (4)

- **Projection of a tuple onto a set of attributes:** used to extract specific values of that tuple according to the attributes specified.
- **Notation** : the projection of a set of values t onto a set of attributes X of a relation R is $t_R[X]$ or $t[X]$. If X has an attribute $t_R.X$
- **For example** : given the relationship HOCVIEN with the attribute set HOCVIEN $^+$ ={Mahv,Hoten,Gioitinh,Noisinh,Malop}, containing 3 sets of values hv_1 , hv_2 and hv_3

2.6 Projection (5)

- Projection of 1 tuple onto 1 attribute
  - hv_1 [Hoten] = ('Ha Duy Lap')

	HOCVIEN				
	Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
$hv_1 =$	K1103	Ha Duy Lap	Nam	Nghe An	K11
$hv_2 =$	K1102	Tran Ngoc Han	Nu	Kien Giang	K11
$hv_3 =$	K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11

2.6 Projection (6)

- Projection of a tuple onto a set of attributes
  - attribute set X={Hoten, Gender}
  - $hv_2[X]$ = ('Tran Ngoc Han','Nu')

	HOCVIEN				
	Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
hv_1 =	K1103	Ha Duy Lap	Nam	Nghe An	K11
hv_2 =	K1102	Tran Ngoc Han	Nu	Kien Giang	K11
hv_3 =	K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11

2.7 Key

2.7.1 Super key

2.7.2 Key

2.7.3 Primary key

2.7.4 Equivalent Key

2.7.5 Foreign key

2.7.1 Super key (1)

- ***Superkey***: is a subset of attributes of Q^+ whose values can distinguish two different tuples in the same instance of any $\boldsymbol{T}_{Q}$.

  That is:

  $\forall\, t_1, t_2 \in \boldsymbol{T}_Q,\ t_1[K] \neq t_2[K] \Leftrightarrow$ K is a superkey of Q.

- A relation has at least one superkey (Q^+) and can have many superkeys.

2.7.1 Super key (2)

- For example, the super keys of the HOCVIEN relation are:

{Mahv};{Mahv,Hoten};{Hoten};{Noisinh,Hoten}…

HOCVIEN				
Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
K1103	Ha Duy Lap	Male	Nghe An	K11
K1102	Tran Ngoc Han	Nu	Kien Giang	K11
K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11
K1105	Tran Minh Long	Male	Ho Chi Minh City	K11
K1106	Le Nhat Minh	Male	Ho Chi Minh City	K11

2.7.2 Key (1)

Key: K is the key of relation R, satisfying 2 conditions:

- K is a super key.
- K is a *smallest non-empty* superkey (contains the fewest attributes and is non-empty).

  Formally:

  $\neg\exists\, K_1 \subset K,\ K_1 \neq \varnothing$ such that K_1 is a super key.
- An attribute that participates in a key is called *a key attribute*, otherwise it is *a non-key attribute*.

2.7.2 Key (2)

- **For example** : the super keys of the HOCVIEN relation are: {Mahv};{Mahv,Hoten};{Hoten};{Hoten,Gioitinh};{Noisinh,Hoten};{Mahv,Hoten,Gioitinh,Noisinh}…

=> then the key of the HOCVIEN relationship can be {Mahv}; {Hoten}

- **For example** : the key of the relation GIANGDAY (Malop, Mamh, Magv, HocKy, Nam) is K={Malop,Mamh}. The key attribute will be: Mamh,Malop. The non-key attribute will be Magv, HocKy, Nam.

2.7.3 Primary key

- When installed on a specific DBMS, if a relation has **more than one *key***, we can only choose one and call it ***the primary key.***
- **Symbol** : *Attributes in the primary key* when listed in a relation must be <u>underlined</u> .
- **For example** :
  - HOCVIEN (<u>Mahv</u>, Hoten, Gioitinh, Noisinh, Malop)
  - GIANGDAY (<u>Mamh, Malop</u>, Magv, Hocky, Nam)

2.7.4 Equivalent Key

- The remaining keys (not selected as primary keys) are called equivalent keys.
- **For example,** in two keys {Mahv},{Hoten}, the primary key is {Mahv}, the equivalent key is {Hoten}

2.7.5 Foreign keys (1)

- Given relations R(U), S(V). $K_1 \subseteq U$ is the primary key of R, and some attributes $K_2 \subseteq V$
- We say **K_2 is a foreign key** of *S* referencing *to* the primary key K_1 of R if the following conditions are satisfied:
  - K_1 and K_2 have the same number of attributes and their semantics are also the same.
  - Between R and S there exists a 1-n relationship on K_1 and K_2.
  - $\forall\ s \in S,\ !\exists\ r \in R$ such that $r.K_1 = s.K_2$

2.7.5 Foreign keys (2)

- For example, given two relations

  LOP (<u>Malop</u>, Tenlop, Siso, Khoahoc)

  HOCVIEN (<u>Mahv</u>, Hoten, Gioitinh, Noisinh, Malop)

- The Malop attribute in the LOP relation is the primary key of the LOP relation. The *Malop attribute in the HOCVIEN relation is a foreign key, referencing Malop in the LOP relation.*

2.7.5 Foreign keys (3)

HOCVIEN				
Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
K1103	Ha Duy Lap	Nam	Nghe An	K11
K1102	Tran Ngoc Han	Nu	Kien Giang	K11
K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11
K1105	Tran Minh Long	Nam	TpHCM	K11
K1106	Le Nhat Minh	Nam	TpHCM	K11

LOP				
Malop	**Tenlop**	**Trglop**	**Siso**	**Magvcn**
K11	Lop 1 khoa 1	K1106	11	GV07
K12	Lop 2 khoa 1	K1205	12	GV09
K13	Lop 3 khoa 1	K1305	12	GV14

2.8 Relational schema (1)

- ***A relational schema*** is intended ***to describe the structure*** of a relation and ***the relationships*** between the attributes in that relation.
- ***The structure of a relation*** is the set of attributes that make up that relation.
- A relation schema consists of a set of attributes of a relation along with a description that defines the meaning and relationships between the attributes.

2.8 Relational schema (2)

- A relational schema is characterized by:
  - A distinguishing name
  - A finite set of attributes (A_1, …, A_n)
- The notation of the relational schema ***Q*** consisting of ***n*** attributes (A_1, A_2,… A_n) is:
  - $Q(A_1, A_2, ..., A_n)$

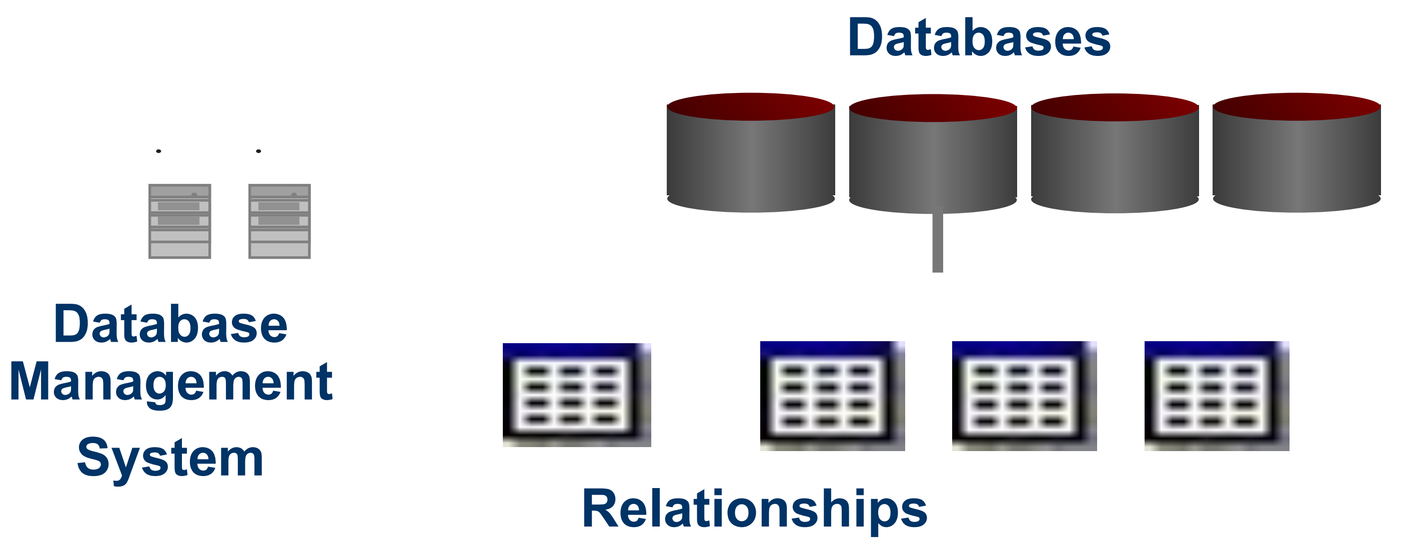
2.8 Relational schema (3)

- HOCVIEN (<u>Mahv</u>,Hoten,Gioitinh,Noisinh,Malop)
- Tân từ: each student has a student code to distinguish them from other students. Need to store their full name, gender, place of birth and class.

HOCVIEN				
Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
K1103	Ha Duy Lap	Male	Nghe An	K11
K1102	Tran Ngoc Han	Nu	Kien Giang	K11
K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11
K1105	Tran Minh Long	Male	Ho Chi Minh City	K11
K1106	Le Nhat Minh	Male	Ho Chi Minh City	K11

2.8 Database Schema (1)

- Is a **set of relational schema** and the relationships between them in the same management system.



"Student Management" Database Schema

HOCVIEN (<u>MAHV</u>, HO, TEN, NGSINH, GIOITINH, NOISINH, MALOP)

Tân từ: mỗi học viên phân biệt với nhau bằng mã học viên, lưu trữ họ tên, ngày sinh, giới tính, nơi sinh, thuộc lớp nào.

LOP (<u>MALOP</u>, TENLOP, TRGLOP, SISO, MAGVCN)

Tân từ: mỗi lớp gồm có mã lớp, tên lớp, học viên làm lớp trưởng của lớp, sỉ số lớp và giáo viên chủ nhiệm.

KHOA (<u>MAKHOA</u>, TENKHOA, NGTLAP, TRGKHOA)

Tân từ: mỗi khoa cần lưu trữ mã khoa, tên khoa, ngày thành lập khoa và trưởng khoa (cũng là một giáo viên thuộc khoa).

MONHOC (<u>MAMH</u>, TENMH, TCLT, TCTH, MAKHOA)

Tân từ: mỗi môn học cần lưu trữ tên môn học, số tín chỉ lý thuyết, số tín chỉ thực hành và khoa nào phụ trách.

DIEUKIEN (<u>MAMH, MAMH_TRUOC</u>)

Tân từ: có những môn học học viên phải có kiến thức từ một số môn học trước.

GIAOVIEN(<u>MAGV</u>,HOTEN,HOCVI,HOCHAM,GIOITINH,NGSINH,NGVL, HESO, MUCLUONG, MAKHOA)

Tân từ: mã giáo viên để phân biệt giữa các giáo viên, cần lưu trữ họ tên, học vị, học hàm, giới tính, ngày sinh, ngày vào làm, hệ số, mức lương và thuộc một khoa.

GIANGDAY(<u>MALOP,MAMH</u>,MAGV,HOCKY, NAM,TUNGAY,DENNGAY)

Tân từ: mỗi học kỳ của năm học sẽ phân công giảng dạy: lớp nào học môn gì do giáo viên nào phụ trách.

KETQUATHI (<u>MAHV, MAMH, LANTHI</u>, NGTHI, DIEM, KQUA)

Tân từ: lưu trữ kết quả thi của học viên: học viên nào thi môn học gì, lần thi thứ mấy, ngày thi là ngày nào, điểm thi bao nhiêu và kết quả là đạt hay không đạt.